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ORIGINAL PAPER

Enhanced uplink handover scheme for improvement of energy efficiency and QoS in LTE-A/5G HetNet with ultra-dense small cells

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Abstract

To construct a femtocell based ultra-dense network (UDN) is the important way to satisfy the ever-growing data traffic demands in the next generation mobile networks. However, it causes frequent handovers (HOs), thus leading to negative consequences such as increased power consumption and poor quality of service (QoS). This paper represents a research on the uplink handover (UL-HO) being watched recently and proposes a new method to further improve the energy efficient and QoS. First, we confirm the UL-HO mechanism, formulate the power consumption caused by UL-HO and compare downlink handover (DL-HO) with UL-HO in terms of the performance parameters including the power consumption, handover rate (HOR), handover failure rate (HOFR), ping-pong rate (PPR), delay, packet loss rate (PLR) and throughput during handover (HO). Then, it is proved through simulation that the reliable target cell determination scheme considering uplink reference signal received power (UL-RSRP), available bandwidth, direction of user equipment (UE), and admission control is lower in HOR, HOFR, PPR, power consumption, delay and PLR. In particular, we propose a new algorithm to determine the reasonable target cell considering UL-RSRP, available bandwidth and moving direction of UE, thus reducing HOR and PPR. The improved UL-HO scheme can be implemented easily by adding some functions besides the upper-layer protocol, while it can provide a lot of benefits such as reduced power consumption, delay, frequent HOs resulting in reducing HOR and PPR. Accordingly, this can be introduced into 5G as a HO scheme.

Keywords Uplink handover · Quality of service · Handover · Femtocell · Small cell

Abbreviations

BBU	Base band unit
BS	Base station
CAC	Call admission control
DL-HO	Downlink handover
DL RS	Downlink reference signal
GBR	Guaranteed bit rate
HeNB	Home eNodeB
HetNet	Heterogeneous network
HCP	HO control parameter
HM	Hysteresis margin
HO	Handover
HOcmd	Handover command
HOconf	Handover confirmation
HOFR	Handover failure rate
HOR	Handover rate
LABRBHA	LTE available bandwidth and RSRP based handover algorithm
MeasReport	Measurement report
MME	Mobility management entity
MR	Measurement report
PLR	Packet loss rate
PPR	Ping-Pong rate
RACH	Random access channel
RSRP	Reference signal received power
RSRQ	Reference signal received quality
SON	Self-organizing network
SRS	Sounding reference signal
TB	Transport block
TTT	Time-to-trigger
UDN	Ultra-dense network
UE	User equipment
UL-HO	Uplink handover
UL RS	Uplink reference signal
UL-RSRP	Uplink reference signal received power
VPT	Virtual pass time
QoS	Quality of service

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1 Introduction

As mobile networks develop, a number of microcells are being overlaid with macrocells to meet the demand for growing traffic in the hot spots. In particular, as the ultra-dense network is formed and the heterogeneous network consisting of cells with different service areas is emerged in LTE-A/5G respectively, the resource allocation and handover become more complex. In [1], the authors proposed a scheme for resource reservation of UEs to mitigate interference in a small cell network, unlike the traditional ones in which both resource allocation and reservation are done at the base station. Here, the UE transmits a resource allocation request message to the base station to reserve resources or reallocate resource blocks, and then a distributed resource block scheme is designed using a Nash bargaining resource allocation algorithm. Finally, it is formulated as an integer optimization problem with relatively small scale, which can be solved by the branch-and-bound method. This shows that UE can affect the key procedures such as resource allocation.

The authors in [2] proposed a method for downlink wireless resource management in a small cell network, which not only avoids the inter-femtocell interference by allocating the same resource blocks to the UE but also enhances the QoS. Then, they set up the optimization problem in terms of improving the inter-cell fairness and maximizing the efficiency of the PRB, and studied the solution to solve the NP-complete problem with the greedy algorithm. This means that the procedure and mechanisms become more complicated due to the simultaneous action of several factors such as the inter-small cell interference and QoS in cellular networks consisting of a large number of small cells. In [3], a resource and power allocation scheme to improve energy efficiency and spectral efficiency in heterogeneous network is proposed. Energy efficiency and spectral efficiency are other important issues in HetNet where macrocells and small cells coexist. As the number of cells and subscribers increase rapidly, energy consumption of the network gets bigger, and it is important to increase the data rate while reducing energy consumption by improving energy efficiency for green LTE-A/5G networks. In addition, spectral efficiency should be improved to achieve high data rate using limited spectral resources, and the authors proposed an energy efficiency optimization problem, studied a new subfrequency reuse scheme. Thus, research in improving energy efficiency and QoS becomes an important direction in LTE-A/5G HetNet with ultra-dense small cells.

The HO is an essential process for mobility in cellular networks. HO refers to the procedure for the UE in RRC_CONNECTED state to change its serving cell [4]. In existing works, most HOs use the HO model where the base station (BS) decides HO based on the UE measurement reports of downlink reference signal (DL RS). Henceforth, this model is referred to as a DL-HO model.

Handover plays a very important role in ensuring the quality of communication in mobile networks. In particular, handover occurs very frequently in LTE-A/5G networks with large number of femtocells, and handover rate increases when UEs move at high speed. On the other hand, since complex procedures such as cell identification, access control, cell search, cell selection, and handover decision are performed during handover, the probability of handover errors such as delay, connection failure, and packet loss gets higher [5].

As can be easily expected, the high mobility of the UE results in a very high HOR in dense small cell deployment, and the HO scheme in this deployment should be different from the ones in small cell deployment [6]. In this paper, the authors proposed a scheme to dynamically adjust the hysteresis margin of RSRP and RSRQ on the basis of the speed and location when a high-speed train enters the overlapping region. In [7], they analyzed the effect of delayed channel quality indicators on HO decisions in dense small cell network, and it is confirmed that the probability of handover failure is increased in a dense small cell network, and that HO is very sensitive to user speed and channel state.

Among the many factors that need to be considered in HO, there are exclusive ones, e.g. load balancing and HO optimization [8]. Self-Organizing Networks (SON), introduced to enhance the automatic management of cellular networks, has important functions to balance network load and optimize the handover. However, the authors found that conflict may occur if both functions are to be implemented simultaneously, and they proposed a coordination scheme to solve this problem. Also, there are studies to reduce the handover rate by optimizing call admission control and bandwidth allocation [9]. Thus, how to rationally set many factors affecting on handover [10] is being an important topic in research of HetNet where macrocells coexist with small cells, and is more highlighted in 5G networks such as the heterogeneous cloud small cell networks that combine HetNet and cloud computing [11] and heterogeneous wireless networks that use different wireless access technologies (RATs). Finally, the research to propose a reasonable HO scheme plays a very important role in improving the quality of service and increasing the reliability and stability of communication in LTE-A/5G HetNet.

Due to this importance of HO, we propose a HO method to improve QoS and the HO performance while reducing energy consumption in LTE-A/5G HetNet.

The main contributions of the paper are as follows.

First, a mathematical model to estimate the power consumed during handover is proposed and it is possible to estimate the power consumption in different HO schemes. The power required to send and receive HO-related signaling messages is quantitatively estimated and a total power consumption model is proposed. By proposing a power consumption model, a new energy-efficient HO scheme is proposed, and it is verified through experiments that the proposed one can be used to estimate the power consumption of any HO scheme.

Second, while studying the energy-efficient uplink handover, we propose a reasonable target cell selection scheme considering various factors such as UE mobility, access admission control, available bandwidth allocation, QoS, etc., to improve the handover performance. We propose a new metric value that reflects the velocity and direction of UE, CAC, available bandwidth allocation, and QoS requirements for determining the optimal target cell in many candidate ones, and discuss the usefulness of the proposed scheme by evaluating the computational complexity of target cell determination.

Third, a simulation system implementing the uplink HO scheme is constructed and the performance of the proposed HO scheme is compared with the previous ones. The proposed method is compared in terms of the HO performance and QoS by newly implementing the uplink HO scheme in the LTE-Sim simulator with different parameters.

This article is organized as follows.

In Sect. 2, a brief literature survey of HO management techniques is presented. Section 3 presents the system model of our approach, and Sect. 4 presents the energy consumption model for uplink handover and presents an algorithm for finding a reasonable target cell by combining the velocity and direction of the UE, bandwidth allocation and call admission control in different scenarios of ultra-dense small cell. Section 5 details the parameters and simulation environments used in the simulation and gives the simulation results. In this chapter, we conduct a performance comparison analysis with previous methods on various indicators to evaluate the effectiveness and usefulness of the proposed method.

2 Related work

As mobile communication develops, data throughput is increasing rapidly [12], and femtocell ultra-dense network technology is being introduced to meet the demand for high data rate in cellular networks [13, 14]. This technique improves the spectral efficiency of cellular networks by reducing the number of UEs each Home eNodeB (HeNB) serves and increasing the frequency reuse rate [13].

However, the ultra-densification of small cells causes frequent HOs, increases signaling overhead and power consumption during HO [15]. Increasing of signaling overhead results in decrease of throughput of UEs [16] and fall of energy efficiency due to the high-power consumption of communication unrelated to data stream. In addition, the frequent HOs increase the probability of failure of HO and in case of several causes of failure of signaling messages during handover, system delay increases and leads to packet loss, and so on, the quality of service (QoS) deteriorates [16]. Therefore, it is important to study the handover in next generation mobile networks (NR) that constitute an ultra-dense cellular network with a small cell.

In [17], on the basis of study on which the optimization of HO may causes negative effect on load balance, a tradeoff method is proposed to achieve a compromise between them. It says that HO is closely related to the cell load. Hence a work to reduce HOR by combining HO, bandwidth allocation, call admission control (CAC) are introduced [18]. What is important in HO is to find a reliable target base station (BS). If the BS selected by HO algorithm is overloaded, the target base station cannot allocate resources such as bandwidth to UE that is handover and entered into its service area, resulting in call failure by the call admission control of the target base station. This causes the UE to disconnect call, so that the handover fails or it need to find a new target cell. This will eventually lead to frequent handover. To overcome this, the authors of the paper provided new method to decide about reliable target BS by combining bandwidth allocation and call admission control (CAC) to reduce the HO rate.

In [19], a method is proposed to optimize the handover while considering load balancing using fuzzy and reinforcement learning. Such an approach can be closely combined with self-organization network techniques (SON). In [20], it was shown that the speed of UE and minimization of latency are an important factor to be considered during HO. Hence many papers have been published that have investigated handover to minimize delay during HO. In [21], a multi-layer neural network was used to select the target BS considering the direction of UE movement to reduce the handover latency during HO. The reduction of the handoff delay means the reduction of packet loss, so we also reduced the packet loss rate together. In [22], the speed and direction of the mobile station are also considered in deciding the handover. Raghavi S in his paper [23] addresses the 'cell dragging effect' and the 'pilot pollution' problem, and to reduce this, they proposed a handover algorithm based on dwell time and movement direction. 'Cell dragging effect' is occurred when UE being still serviced by serving cell enters into the coverage of target cell, and still the UE receives better RSRP from serving cell. 'Pilot pollution' indicates that at a location,

the power of signal transmitted from several cell is observed at the UE, but it is difficult to specify power of signal of which cell is dominant. It is shown in this paper that predictions of the direction of travel and velocity are helpful in selecting the optimal target subregion in this environment.

Handover consumes energy during the sending and receiving of messages for it, and frequent handover results in increased energy consumption. Many studies have been carried out to reduce the energy dissipated in HO. In [24], a method to reduce the frequency of too early handover by extending the bandwidth to reduce the energy consumption in LTE is proposed.

In some works, such as [25], regression model which decide the HO control parameters (HCP) such as Hysteresis Margin (HM), A3Offset, time-to-trigger (TTT) dynamically is proposed to control HO by the configuration of HCP. Similarly, Abdulraqeb Alhammadi [26] also studied on the reduction of Handover Failure Rate (HOFR) by dynamically calibrating HCPs in LTE-A/5G network. In this paper, a new algorithm for dynamically setting the HCPs in HetNet, which consists of LTE-A and millimeter-wave networks, is proposed to compare the characteristics at different velocities of the UE. It is studied to improve HO performance considering several parameters used for HO. In [27], the authors considered RSRP and available bandwidth simultaneously, provided new HO algorithm and the its performance is simulated and evaluated by using open source simulator LTE-Sim.

One common point of all these studies is to decide to HO based on the measurement of downlink reference signal received strength at the UE, i.e. if UE received DL RS sends a measurement report message to the base station, HO is performed at base station by deciding HO on the basis of the report and sending HO command to UE, consecutively.

However, recently, unlike such traditional schemes (which is hereafter referred to as downlink handover: DL-HO), the studies on new uplink handover (UL-HO) schemes using uplink reference signal (UL RS) have been reported [28–30]. These papers have proposed a handover scheme based on uplink reference signals, based on estimating the step-by-step power consumption in the downlink handover strategy and revealing that the measurement report messages are the largest contributor to power consuming. In the scheme, UE sends UL RS to BS, then BS measures RSRP of UL RS and takes HO decision. This method omits one procedure, such as the Measurement Report Request message, since when a UE transmits an uplink reference signal to the base station, it determines a handover by measuring the reference signal strength at the base station, thus reducing the power consumed in the handover. This approach is based on adding some simple functions to the 3GPP standard, while gaining benefits from the energy consumption point of view.

The X2 interface in LTE-A network fulfills important function in optimization of HO management. In [31], to evaluate the effectiveness of using X2 HO and S1 HO, authors proposed HO parameters for the services with different QCIs, e.g., HTTP Web Browsing, video streaming, audio traffic and estimated the effect on whole network performance. It is proved that X2 interface has more effect on HO.

In [32] and [33], authors have studied on HO management scheme in Femtocell-Macro cell network considering admission control, moving direction and velocity of UE to address frequent HOs in LTE-A HetNet. They proposed a channel borrowing admission control scheme to optimally adjust the handover using admission control, and consider other parameters such as UE velocity and direction to effectively reduce the undesired handover.

Through the analysis of aforementioned researches, we can see that most studies only discuss DL-HO within the framework of 3GPP standard, and only a few studies discuss UL-HO. However, UL-HO is very attractive due to the advantage such as decreasing of HO delay due to lower power consumption and simplification of procedure. Therefore, we conclude that in reducing unnecessary handover and reducing handover delay and failure rate, several factors such as the direction and speed of the mobile station's movement, access permission control, and the presence or absence of available bandwidth work together, but only some of them are considered in many papers. The HO management problem is more problematic in heterogeneous networks (HetNet) where the small cell is ultra-dense and mixed with macro base stations, because the small cell handover rate (HOR) is very high when moving UEs in such environments.

Hence in this research work, we solve the following problem:

First, a power consumption model considering the transmit power of the reference signals is proposed and based on this, the energy consumption of the uplink handover and downlink handover models is compared. Yet, there are few studies in previous studies.

Second, we develop an algorithm to select a reasonable target cell considering the velocity and direction of movement of the UE, available bandwidth and call admission control in uplink handover. Choosing the optimal target cell with respect to UE mobility, available bandwidth, and call admission control is of great importance in reducing frequent HO in cellular networks with ultra-dense small cell. In previous studies, there have been few studies on reducing the HOR by combining three factors rationally, especially in relation with uplink HO.

Third, we verify the effectiveness and usefulness of the proposed method through a comparative analysis of various indices between the proposed method and the previous method.

3 System model and formulation

In this chapter, we first propose a system model of UL-HO using UL RS (Sounding Reference Signal (SRS)) and a mathematical model to evaluate the total energy consumed during HO, and then present our proposed HO framework based on search of the optimal target cell.

3.1 Uplink handover (UL-HO) model

In general, measuring RSRP for downlink reference signals and determining handover based on it is the downlink handover that has been used so far. However, in the uplink handover model, we determine the handover based on measuring the RSRP of the uplink reference signal, Sounding RS.

Figure 1 shows the proposed UL-HO system model in this paper. As shown in the figure, there are small cells around the macro cell, and this network architecture is called a heterogeneous network, HetNet. Unlike prior DL-HO, the serving BS measures RSRP of UL-RS, i.e. SRS and transmits HANDOVER_REQUIRED message to Mobility Management Entity (MME) through S1 interface upon fulfilling the condition where RSRP of UL-RS (UL-RSRP) is lower than threshold. In response, MME sends HANDOVER_CAMMAND to serving BS [34]. The serving BS sends HO request to target BS through MME and target BS prepares for HO, returns HO request acknowledgement. The major difference compared to the DL-HO procedure is that the UL-HO eliminates the measurement control message and, in particular, the measurement report message. This can be an important benefit in terms of energy consumption and HO latency.

In this paper, power consumption during HO is formulated to evaluate the saved energy due to simplification of procedure, and it is described in chapter 4.

3.2 Framework to choose the suitable target cell

In this research work, new framework to choose the optimal target cell is proposed. In order to reduce HOR and HOFR, decrease PLR and HO delay during HO, it is very important to select the optimal target cell.

First, to choose the optimal target cell, Virtual Pass Time (VPT) considering the average velocity vector and the coverage of cell is defined and the candidate macrocell or small cells are ranked in terms of VPT. As VPT reduces frequent HOs and HOR simultaneously, it is particular

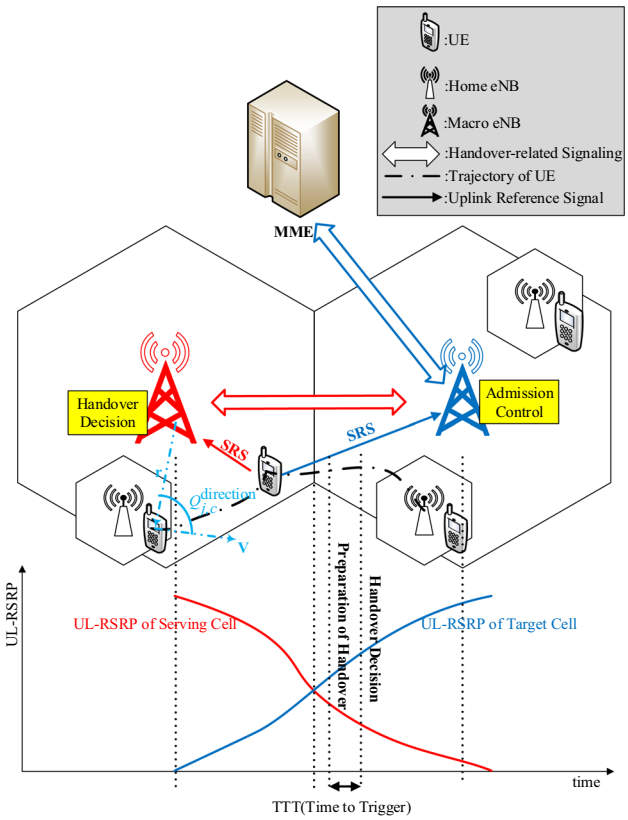

Fig. 1 Uplink Handover (UL-HO) system model (**r**: radial vector, **v**: velocity vector, $Q_{j,c}^{direction}$: angle between **r** and **v** for j th UE that is served by cell c, *MME* Mobility Management Equipment, *UL-RSRP* Uplink Reference Signal Received Power)

interest in HetNet where the coverage of cell and UE's velocity are different. This is because the lower HOR, the longer the average time to stay in the cell.

Next, we define new metric which considers three parameters, i.e. available bandwidth B_a, UL-RSRP, direction of UE's velocity vector and propose the framework to choose the suboptimal target cell on the basis of that metric. Yet, the metric values considering these three parameters have not been considered, especially for UL-HO, and the results of the proposed metric are not reported. In particular, we further improved the handover performance by changing the auxiliary parameters that are input to the metric function according to the type of service.

Figure 2 shows the proposed target cell select framework.

First, we input data on the velocity and direction of the UE and cell, while we compute VPT proposed in chapter 4, and then order the candidate cells accordingly, and enter the target cell search which will start from the candidate cell with the longest VPT.

Next, we determine the type of service of UE and the auxiliary parameters α, β, γ of the metric function discussed in Chapter 4, depending on whether it is a GBR

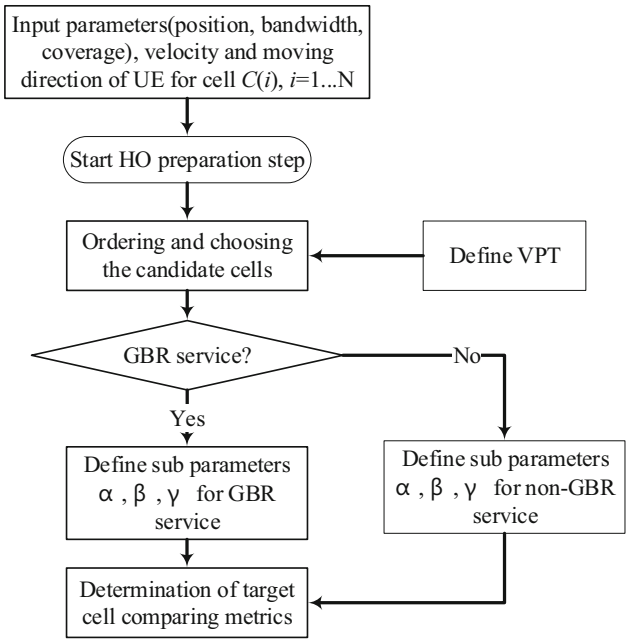

Fig. 2 Proposed framework to choose the target cell

service or a non-GBR service. These values are different according to the type of service. Based on this, the metric is determined and the optimal target cell is finally identified.

The effectiveness of the proposed method is verified through simulation in Sect. 5.

4 Power consumption during UL-HO and choosing reliable target cell

In this chapter, we first propose an algorithm to obtain an optimal target cell based on mathematical modeling of the energy consumed in UL-HO and defining a VPT and a new metric to estimate how much energy is consumed during UL-HO and how much energy this scheme saves compared to DL-HO.

4.1 UL-HO power consumption

In this section, we present a power consumption model that calculates the power consumed for the transmission and reception of various signaling messages during UL-HO. Using the power consumption model, we calculate how much power is required by the Base Station (BS) or UE to transmit and receive HO signaling messages.

4.1.1 Power consumption model at transmitter

Signaling messages which occur during UL-HO are "HOcmd" (Handover Command) transmitted from the BS and "RACH", "HOconf" (Handover confirmation) transmitted from the UE. When the size of signaling message is $\Lambda_x(\mathrm{x} = \{RACH, HOcmd, HOconf\})$ and the number of information bits per Transport Block (TB) $\Lambda_{TB} = 152bits$ [35], the required number of TBs for each signaling message x, N^x_{TB} can be obtained by:

$$N^x_{TB} = \frac{\Lambda_x}{\Lambda_{TB}}, \quad x = \{MR, HOcmd, HOconf\} \tag{1}$$

where Θ denotes the smallest integer larger than or equal to Θ.

From the 3GPP standard [36], the allocated power per TB at the BS and UE can be formulated as:

$$P^{TB}_{BS} = P_{BS}/\mathrm{M}^{DL}_{TB} \tag{2}$$

$$P^{TB}_{UE} = P_{UE}/\mathrm{M}^{UL}_{TB} \tag{3}$$

where M^{DL}_{TB} (M^{UL}_{TB}) is the number of TBs in the DL (UL) given by a system bandwidth B_s.

The allocated transmitted power (in W) per signaling message x is:

$$P^x_{node,Tx} = P^{TB}_{node} \cdot N^x_{TB}, node = \{BS, UE\} \tag{4}$$

Now, some well-known power consumption models [37–39] are applied and the power consumption for the BS transmitting signaling message x, $P^{x,Tx}_{BS,sup}$ is written as:

$$P^{x,Tx}_{BS,sup} = P^x_{BS,Tx}/\kappa + N^x_{TB}/M^{DL}_{TB} \cdot \left(P_{RF,BS} + P^{BP}_{BS}\right) \tag{5}$$

where $P^x_{BS,Tx}$ is the transmitted power given by (4) and κ is the power amplifier efficiency. $P_{RF,BS}$ denotes the supply power for the RF equipment and P^{BP}_{BS} is the power consumed in the Base Band Processing Unit (BP) of BS in W (see Table 1). Similarly, the power consumption for the UE to transmit a signaling message x is given by:

$$P^{x,Tx}_{UE,sup} = P^x_{UE,Tx} + N^x_{TB}/M^{UL}_{TB} \cdot \left(P_{RF,UE} + P^{BP}_{UE}\right) \tag{6}$$

where $P^x_{UE,Tx}$ is the transmitted power given by (4) and P^{BP}_{UE} is the power consumed in UE BP (Base Band Processing Unit) in the case of transmission from UE (see Table 1).

Now, the time-averaged power consumption in network nodes $(node = \{BS, UE\})$ is defined in (7).

$$\overline{P}^{x,Tx}_{node,sup} = P^{x,Tx}_{node,sup} \cdot \left(\Delta t^x_{node}/\Delta\tau\right) \tag{7}$$

where $P^{x,Tx}_{node,sup}(node = \{BS, UE\})$ is the power defined in (5)-(6), and $\left(\Delta t^x_{node}/\Delta\tau\right)$ represents the percentage of time where the signaling is actually transmitted. Assuming that signaling message x has a duration of T^x_{node} seconds (see Table 1) and a number of K_x signaling messages are transmitted over a period of $\Delta\tau$ in seconds, obtaining $\Delta t^x_{node} = T^x_{node} \cdot K_x$. Finally, we can rewrite (7) as follows.

$$\begin{aligned}\overline{P}^{x,Tx}_{node,sup} &= P^{x,Tx}_{node,sup} \cdot T^x_{node} \cdot (K_x/\Delta\tau) \\ &= P^{x,Tx}_{node,sup} \cdot T^x_{node} \cdot R^x_{node}\end{aligned} \tag{8}$$

Table 1 Power consumption parameters and values

Feature	Values
Signaling message sizes	Λ_{MR} = 128bits; Λ_{HOcmd} = 296bits; Λ_{HOconf} = 96bits [40]
Carried bits in a TB	Λ_{TB} = 152bits (QPSK, Λ_{RE} = 2) [35]
Number of TBs per signaling message	N_{TB}^{MR} = 1, N_{TB}^{HOconf} = 1, N_{TB}^{HOcmd} = 2, N_{TB}^{RACH} = 1
Power amplifier efficiency	κ = 0.311(31.1%) [37]
RF supply power	$P_{RF,BS}$ = 12.9W [27], $P_{RF,UE}$ = 2.35W [38]
BS BP supply power	P_{BS}^{BP} = 29.4W [39]
UE BP supply power	P_{UE}^{BP} = 0.62mW [38]
Received BP power (mW)	$P_{RxBP}(R_{Rx}) = 0.97.R_{Rx} + 8.16$ [38]
Signaling transmission times	T_{BS}^{HOcmd} = 1 ms; T_{UE}^{MR} = 1 ms; T_{UE}^{HOconf} = 1 ms; T_{UE}^{RACH} = 1 ms [41]

where $R_{node}^{x} \cdot (K_x/\Delta\tau)$ is defined as the signaling rate which is the number of signaling messages transmitted per unit time. The main numerical parameters are shown in Table 1.

The supply power for the BS and UE to transmit signaling message is the main basic data for evaluating power consumption during HO.

4.1.2 Power consumption model at receiver

In this paper, the power consumption to receive signaling message is calculated as below. The received power (in W) per signaling message is obtained according to the following expression:

$$P_{node,Rx}^{x} = P_{node,Tx}^{x} \cdot G_{i,j} \cdot |h|^2 + P_n^x, \quad node = \{BS, UE\} \tag{9}$$

where $G_{i,j}$ is the total path gain for i^{th} UE and j^{th} BS (including distance-dependent attenuation, the angular antenna gain, and shadow shading), $|h|^2$ is the fast fading contribution and P_n^x is the additive white Gaussian noise contribution over the bandwidth of the signaling message x.

The power required to receive and process the signaling message x in the BS and the UE, $P_{node,sup}^{x,Rx}, node = \{BS, UE\}$ can be written as below according to [35–39].

$$P_{BS,sup}^{x,Rx} = P_{BS,Rx}^{x}/\kappa + N_{TB}^{x}/M_{TB}^{DL} \cdot \left(P_{RF,BS} + P_{BS}^{BP}\right) \tag{10}$$

where $P_{BS,Rx}^{x}$ is the received power given by (9) and κ is the power amplifier efficiency. Similarly, the power consumption for the UE to receive the signaling message x is given by:

$$P_{UE,sup}^{x,Rx} = P_{UE,Rx}^{x} + N_{TB}^{x}/M_{TB}^{UL} \cdot \left(P_{RF,UE} + P_{RxBP}(R_{Rx})\right) \tag{11}$$

where $P_{UE,Rx}^{x}$ is the received power given by (9), P_{RxBP} is the received BP power (see Table 1) and R_{Rx} is the received data rate that can be found as $R_{node}^{x} \cdot \Lambda_{TB}$.

Thus, the time-averaged power consumption can be written as:

$$\begin{aligned}\overline{P}_{node,sup}^{x,Rx} &= P_{node,sup}^{x,Rx} \cdot T_{node}^{x} \cdot (K_x/\Delta\tau) \\ &= P_{node,sup}^{x,Rx} \cdot T_{node}^{x} \cdot R_{node}^{x}\end{aligned} \tag{12}$$

Finally, the total supply power for the network is defined as:

$$P_{net}^{total} = \sum_{\Omega_{UE}} \left(\overline{P}_{UE,sup}^{x,Tx} + \overline{P}_{UE,sup}^{x,Rx}\right) + \sum_{\Omega_{BS}} \left(\overline{P}_{BS,sup}^{x,Tx} + \overline{P}_{BS,sup}^{x,Rx}\right) \tag{13}$$

where Ω_{node} is the set of network nodes $node = \{UE, BS\}$.

4.2 Defining VPT and VPT-based confirming the scope of candidate cells

In the works about HO in HetNet [42], a scheme was proposed to allow fast UEs to handover to macro cell and slow UEs to handover to small cell to prevent the frequent HOs when a UE with high mobility passes through a cell. This is the method to increase the residence time in the serving cell.

To generalize and evaluate this scheme better, we define the VPT as follows. For the average velocity vector of j th UE $\overline{\mathbb{v}_j}$ and the area of cell c $\mathcal{S}_c$, VPT $\ell_{j,c}$ is defined as:

$$\ell_{j,c} = \varrho \cdot \frac{\sqrt{\mathcal{S}_c}}{\left|\overline{\mathbb{v}_j}\right|} \tag{14}$$

where ϱ is the correcting parameter defined by the type of cell and is larger than 0, smaller than 1. Also, $|\,|$ denotes the absolute value of a vector. VPT given by (14) is in the unit of time and related to the time where the UE passes by the cell. Therefore, the long VPT means the long time to pass through the cell. To choose a target cell among the ones with high passing time means reducing frequent HOs. Hence, we calculate the VPT of candidate cells including

macrocells and small cells, aligned the candidate cell in a large order, then applied the target cell determination scheme to determine the optimal target cell.

4.3 Determination of the optimal target cell based on new metric

In this article, the new metric is proposed to estimate the optimal target cell. Main factors in estimating the target cell are RSRP, available bandwidth used for admission control and moving direction of UE. However, these three factors are independent of each other and each parameter has the different weight (α, β, γ) according to its effect on HO decision.

In this article, the new metric is proposed to estimate the optimal target cell. Main factors in estimating the target cell are RSRP, available bandwidth used for admission control and moving direction of UE. However, these three factors are independent of each other and each parameter has the different weight () according to its effect on HO decision.

The proposed metric for j th UE and BS c is shown as:

$$\mathcal{M}_{j,c} = \alpha \cdot \mathcal{P}_{j,c}^{UL-RS} + \beta \cdot \left(\frac{\mathcal{B}_c^{ava-Band}}{\mathcal{B}_0}\right) + \gamma \cdot \mathcal{Q}_{j,c}^{direction}, \left(j \in \Omega_{UE}, c \in \Omega_{BS}^{VPT}\right) \quad (15)$$

where the relation between the parameters are given as:

$$\alpha + \beta + \gamma = 1 \quad (0 < \alpha < 1, 0 < \beta < 1, 0 < \gamma < 1) \quad (16)$$

where Ω_{BS}^{VPT} is the set of candidate cells given by the scheme proposed in Sect. 4.2. $\mathcal{P}_{j,c}^{UL-RS}$ is the power strength received in BS for SRS transmitted by j th UE in unit of dBm, while $\mathcal{B}_c^{ava-Band}$ is the available bandwidth of c th BS ($\mathcal{B}_0 = 1MHz$) and $\mathcal{Q}_{j,c}^{direction}$ is the angle from velocity vector to radial vector in unit of radian(see Fig. 1), and α, β, γ denotes the combination weight.

Thus, each term in (15) is essentially all dimensionless and can be added to each other without conflict. If $\alpha = 1, \beta = 0, \gamma = 0$, (15) implies a metric value that determines the target cell based only on the received power of the uplink reference signal (UL-HO based on UL-RSRP in Chapter 5, a HO algorithm named UL-RSRP in short) and if $\alpha = 0, \beta = 1, \gamma = 0$, it means a metric value that determines the target cell based only on the available bandwidth. Similarly, in case of $\alpha = 0, \beta = 0, \gamma = 1$, it will only mean the metric value that determines the target cell based on the moving direction of UE. Hence, the weights α, β, γ are the indicators of the received power, available bandwidth, UE's moving direction, and the degree to which of these three factors should be considered in determining the target cell, respectively, thus the set of appropriate values of α, β, γ means that the three factors can be accurately considered to determine the target cell by considering the influence of the handover. In other word, first term in (15) represents the contribution of the reception power of SRS transmitted from the j th UE to the c th BS, while second term denotes the contribution of the available bandwidth for the j th UE in the c th BS in the metric for determining a target cell. Third term is the metric which shows whether the j th UE is approaching to the c th BS or not. Here, the c th BS includes the serving BS and the candidate BS during HO. While the radial vector between the j th UE and the c th BS is $\boldsymbol{r}_{j,c}$, the angle from the axis x to the instantaneous velocity vector of UE $\boldsymbol{v}$ is φ, third term $\mathcal{Q}_{j,c}^{direction}$ can be written as:

$$\mathcal{Q}_{j,c}^{direction} = -\left|2 \cdot \left(\varphi - \arctan\frac{r_{j,c}^y}{r_{j,c}^x}\right)\right|^2 \quad (17)$$

where $\boldsymbol{r}_{j,c} = r_{j,c}^x \cdot \vec{i} + r_{j,c}^y \cdot \vec{j}$, $\vec{i}, \vec{j}$ is the unit vector of x axis and y axis, respectively. While an RB has a bandwidth of 180 kHz, $\mathcal{B}_c^{ava-Band}$ can be calculated by subtracting the allocated RBs from the total RBs of system and multiplying by 0.18 to obtain the available bandwidth in MHz. As can be seen from Eq. (15), for the cth BS, the larger the value of UL-RSRP that the base station is receiving, the larger the redundant bandwidth of that base station, and the more likely the mobile station is moving towards that base station, the greater the probability that the cth base station will become the target base station.

In the proposed scheme, MME calculates the $\mathcal{M}_{j,serving}$ of serving eNodeB which UE j is attached and compares with $\mathcal{M}_{j,candidate}$ of the candidate BSs in the way in Sect. 4.2.

Finally, the reliable target cell which be chosen for j^{th} UE is defines as:

$$T_{BS(j)} = \underset{c}{\operatorname{argmax}}\, \mathcal{M}_{j,c}, \forall j \in \Omega_{UE}, c \in \Omega_{candidateBS} \quad (18)$$

where $\Omega_{candidateBS}$ is the set of the decided candidate BSs by the way in Sect. 4.2.

4.4 The impact of the service type on deciding the target cell

We propose a new metric based on Eq. (15) to modify the weights by considering the service type. In general, non-GBR services are not more demanding than GBR services in terms of bandwidth. If we assign uniform weights of available bandwidth for GBR and non-GBR services, it is possible that, in the case of non-GBR services, unnecessary handover and frequent HOs due to high weight of available bandwidth can occur, in spite of serving BS can offer enough service, resulting in high Handover Rate (HOR) and Ping Pong Rate (PPR). This requires the adjustment of

Table 2 The reliable HO algorithm

1.	Initialize the parameters of BSs and UEs.
2	If service type of UE j = GBR then
3	$\alpha = \alpha_{GBR}, \beta = \beta_{GBR}, \gamma = \gamma_{GBR}$
4	else
5	$\alpha = \alpha_{non-GBR} = \alpha_{GBR}, \beta = \beta_{non-GBR}, \gamma = \gamma_{non-GBR} = \gamma_{GBR}$
6	End if
7	Calculate $\mathcal{P}_{j,serving}^{UL-RS}$(W), $\mathcal{B}_{serving}^{ava-Band}$(MHz), $\mathcal{Q}_{j,serving}^{direction}$(rad) for serving eNodeB
8	Calculate the metric of serving BS $\mathcal{M}_{j,serving}$ using (15)
9	For $c \in \Omega_{BS}$
10	Calculate the VPT $t_{j,c}$
11	End For
12	Order the BSs according to $t_{j,c}$
13	Determine the set of candidate BSs$\Omega_{candidateBS}$
14	For $c \in \Omega_{candidateBS}$
15	Calculate $\mathcal{M}_{j,c}$ using (15)
16	If $\mathcal{M}_{j,c} > \mathcal{M}_{jmax}$ then
17	$\mathcal{M}_{jmax} = \mathcal{M}_{j,c}$
18	End If
19	End For
20	If $\mathcal{M}_{jmax} > \mathcal{M}_{j,serving}$ then
21	Determine the target BS $T_{BS(j)}$ using (18). HO execution.
22	End If

the weight of available bandwidth according to the service type.

So, in this work, β is defined differently according to the service type:

$$\left.\begin{array}{l}\alpha_{GBR}+\beta_{GBR}+\gamma_{GBR}=1,\ for\,GBR\,service\\ \alpha_{non-GBR}=\alpha_{GBR},\gamma_{non-GBR}=\gamma_{GBR},\beta_{non-GBR}\neq\beta_{GBR},\ for\,non-GBR service\end{array}\right\} \tag{19}$$

All appropriate weights are determined by simulation. It should be emphasized here that (15), (16) and (19) are inseparable from each other, as can be seen from the derivation of (19). In other words, (19) means that we can reasonably determine the values of α, β, γ satisfying (16) for GBR services, and then only the values of β are modified accordingly for non-GBR services. This is because, as explained earlier, the requirement for available bandwidth in the case of non-GBR services is significantly different from that of GBR services.

In fact, the metric value $\mathcal{M}_{j,c}$ of (15) has no physical meaning, and only the relative importance of the metric values is significant. Consequently, among candidate cells, the probability that a cell with good channel conditions will be a target cell should be high, which is reflected in the first term of (15). Also, a cell with a large available bandwidth among the candidate cells should be the target cell, which can increase the HO success rate, which is reflected in the second term. The cell in the moving direction of the UE must be defined as the target cell, which reduces the probability of unsuitable HO such as the ping-pong effect, which is reflected in the third term. It is the weights α, β, γ that reflect which of these three

factors is the larger specific weight assigned to which, and similar to Back-Propagation Learning neural networks [43], the weights can be obtained optimally through experiments, although the three factors that are combined contain different physical contents and range of value changes are different.

4.5 The algorithm of the proposed scheme

The proposed HO algorithm is shown in Table 2.

As easily seen from the table, the computational complexity of the proposed algorithm is $O(\mathrm{C})$ when the number of the candidate cells is C, and thus the same as other HO schemes, i.e. the LABRBHA and UL-RSRP. In other words, the proposed method achieves better performance compared to the previous ones without more complicated computations.

5 Simulation analysis

5.1 Simulation model and parameters

The proposed HO algorithm is simulated to evaluate the usefulness. We have used the open source LTE-Sim [44] to simulate, in the results of the calculations obtained, only the bar diagram was obtained using Excel.

All simulation model parameters are summarized in Table 3 [27, 30]. The location of the base station is fixed, the placement of the UEs follows a random uniform distribution, and the moving direction of the UEs is also randomly set. The location of the base station is fixed, the placement of the UEs follows a random uniform distribution, and the moving direction of the UEs is also randomly set.

Table 3 The values of parameters used in the simulations

Parameters	Values
Frame structure	FDD
Network topology	A hexagonal grid of 10 macrocells and 4 small cells around each macrocell
Radius and area of macro cell	500 m, 0.785 km^2
Radius and area of small cell	20 m, 0.001256 km^2
System bandwidth B_s	5 MHz(25RBs)
Number of RBs per TB	6RBs
M_{TB}^{DL}, M_{TB}^{UL}	25/6
Carrier frequency	f_c = 2.1 GHz
BS DL power	43dBm
UE Power	23dBm
Channel model	Macro Cell Urban
Shadow fading	Log-normal shadowing mean 0 dB, Standard deviation:6 dB
Propagation model	$128.1 + 27.6\log_{10}(R)$, R in 1 km
Velocity of UE	150 km/h, 30 km/h, 6 km/h
Number of UEs	50–150
Max delay	0.1 s
VIDEO bit rate	242Kbps
VOIP bit rate	8.8Kbps
HO parameters	Time to trigger (TTT) = 32 ms, A3 offset = 1 dB
SRS periodicity (T_{SRS})	40 ms
DL scheduling algorithm	Proportional fair [45]
ϱ	0.5
Simulation time	40 s

5.2 The performance of UL-HO and DL-HO with respect to handover and power consumption

Firstly, we have compared the handover performance and the power consumption between UL-RSRP based UL-HO (briefly UL-RSRP) and LABRBHA [27] based DL-HO (briefly DL-HO). For this purpose, we first simulated 10 times using the DL-HO scheme based on the LABRBHA to obtain the average values of all performance indices. Then, also the simulation using UL-HO scheme based on UL-RSRP is conducted in the same manner, and the average values of the performance indices are compared.

5.2.1 Handover performance comparison

Figure 3 shows the comparison of the DL-HO and UL-HO in terms of HO performance parameters i.e. HOR, HOFR and PPR. HOR is defines as the number of HOs occurred in the unit time and it is measured in HO events per second. HOFR is defined as the ratio of HO failure events to total HO events. PPR is defined as the number of ping pong events during a period of time. Notably, the UL-HO reduces significantly the HOFR in comparison of DL-HO. This is because UL-HO doesn't need MeasReport message transmission, thus the HO procedure completes before the UE loses its connection to the serving BS. The elimination of the sending and receiving procedures of one message which was essential for handover, reduces handover delay, and decreases handover rate due to the low handover failure rate. Since HO failures should attempt handover to other cells again, the decrease of handover failure rate leads to a decrease of handover rate. Similarly, the ping pong rate is also reduced. This trend is in good agreement with the results in [27–29], which also confirms the correctness of our simulations.

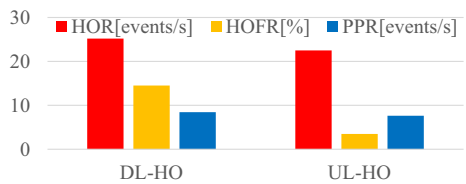

Fig. 3 The comparison of DL and UL-HO scheme in terms of HO performance including handover rate (HOR), handover failure (HOF) and ping-pong rate (PPR) for entire network

5.2.2 Power consumption comparison

We have also compared features of DL-HO and UL-HO in terms of the transmission power consumption. To analyze the energy benefit during UL-HO, only the power consumed due to the HO signaling message transmission in UE and BS is evaluated while UL-HO is triggered by only pure UL RSRP.

For this purpose, the average power consumption per network node is evaluated as follows based on (8), (12), (13).

$$\overline{P}^{Tx}_{node} = \frac{\sum_{j=1}^{N_{node}} \sum_{x} \overline{P}^{x,Tx}_{node_j,sup}}{N_{node}}, \quad \overline{P}^{Rx}_{node} = \frac{\sum_{j=1}^{N_{node}} \sum_{x} \overline{P}^{x,Rx}_{node_j,sup}}{N_{node}}, \quad node = \{UE, eNodeB\} \tag{20}$$

where N_{node} denotes the number of network nodes, i.e. UE or eNodeB and message x is the same from (1).

Now, the UL-HO scheme is called as UL-RSRP.

5.2.2.1 Power consumption for transmission Figure 4 shows the comparison of DL-HO and UL-HO in terms of power consumption when the number of UEs ranges from 50 to 150. In the figure, UE(/eNB) Tx means the average power consumption of UE(/eNodeB) for transmission in DL-HO and UL-HO. UE(/eNB) Rx denotes the average power consumption of UE(/eNodeB) for reception and Network denotes the total power consumed in entire network. As shown in figure, the UL-HO outperforms the DL-HO in terms of power consumption, because UL-HO does not require the MeasReport message transmission. The power consumption of UE for transmission in UL-HO is 48.6% lower than DL-HO in the case when the number of UEs is 100. The significant decrease in power consumption in the UL-HO scheme compared to the DL-HO scheme is due to the absence of the MeasReport message, but as shown in Fig. 3, the reduction in HOR compared to DL-HO is due to the amplification of the effect of not transmitting the MeasReport message. Such an analysis is also well suited to the power consumption estimates described below.

The power consumption of BS for transmission is 11% bigger than UL-HO because of high HOR due to high PPR, as we noted in Fig. 3.

5.2.2.2 Power consumption for reception We have compared power consumption of DL-HO and UL-HO due to the reception of signaling messages. The signaling messages that are transmitted from the UE side are received at the BS and similarly, the messages transmitted from the BS side are received at the UE.

The graph follows the same trend as we noted above (see Fig. 4). The UL-HO is almost 17.4% lower than DL-HO when the number of UEs is 100. Comparing the average power consumption of BS for reception due to the HO-related signaling message reception for DL-HO and UL-HO, that of UL-HO is almost half (47.5%) lower than DL-HO, because UL-HO does not require the 'MeasReport' signaling reception.

5.2.2.3 Total power consumption in network We have compared the total average supply power consumption (i.e. the sum of power consumption for transmission and reception at the both UEs and BSs) of the network for DL-HO with that for UL-HO (see Fig. 4). The graph exhibits the DL-HO consumes more power than UL-HO, it is because of the high HOR and PPR as shown in Fig. 3. The UL-HO gains almost 27.5% benefit of the total power in comparison of DL-HO when the number of UEs is 100.

5.3 availability evaluation of the proposed metric

All the above-mentioned results do not yet include the effectiveness of the new metric proposed in this paper, but refer to only the difference between the pure UL-HO and DL-HO algorithms. In this section, we report the simulation results on the effectiveness of the newly proposed metric.

5.3.1 Estimate the optimal set of parameters α, β, γ according to the service profile

Firstly, we have defined the optimal weights α, β, γ according to the service type by simulation.

We performed simulations on 100 UEs with different services to verify the effectiveness of changing the value of β according to the type of service. Without loss of generality, we randomly set the service type for any UE whether GBR or non-GBR, and once the service type is determined for all UEs, we performed all the HO simulations in case of considering the service type and not, respectively. Then, we evaluated the overall HO performance of the network

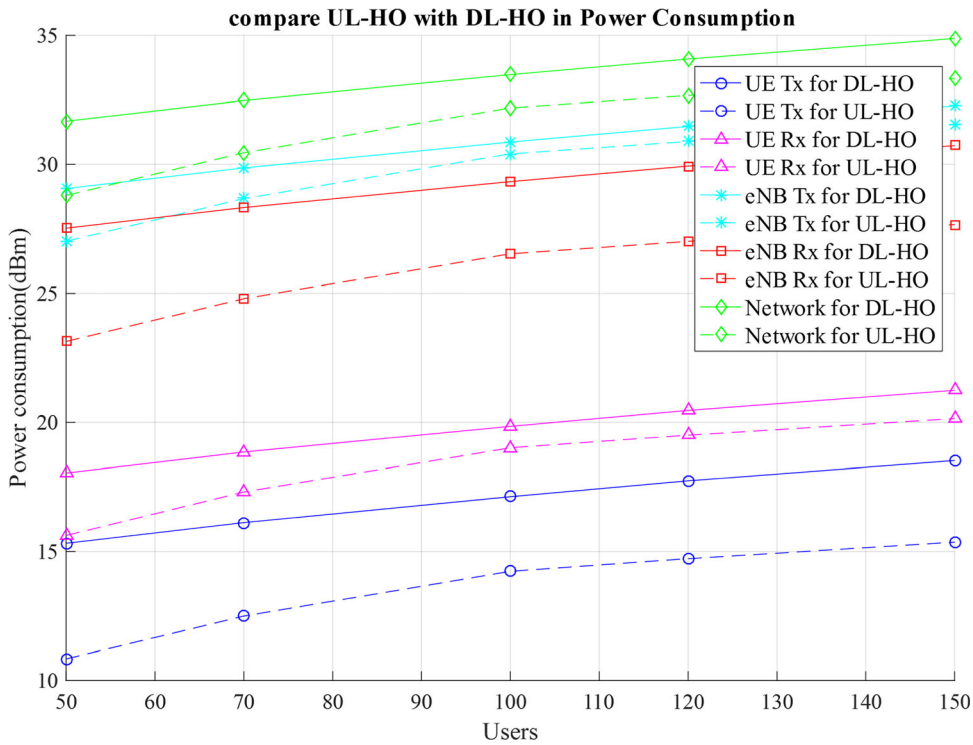

Fig. 4 Comparison of DL-HO and UL-HO in terms of power consumption when transmitting and receiving for UE, eNB and Network

for each method, obtained the average performance value by repeating these experiments 10 times subsequently.

First, we analyzed the effect of randomly setting α and γ values and then changing the β values according to the service type. When α and γ are defined accordingly as $\alpha = 0.8$, $\gamma = 0.1$, $\beta = 0.1$ is given by (19) for GBR service. In the case of Non-GBR service, β was varied from 0.01 to 0.05 to determine the value of β at the highest metric value, while $\beta = 0.01$ was the best. The results are shown in Fig. 5 and 6. Figure 5 a) shows the handover performance of the HO scheme when computing the metric of target cell by assigning the same value of β to all services, regardless of the service type of UE. b) shows the performance of the HO scheme when computing the metric of target cell by assigning a value of 0.1 for GBR service and 0.01 for non-GBR service. As shown in Figures, HOR and PPR is lower due to the decrease of HOR and PPR because the weight is defined less for non-GBR service where requires low bandwidth demand.

However, HOFR is increased because α and γ are not defined properly, thus reduces the compatibility of choosing target cell. This exhibits that the total HO performance is enhanced only if α and γ are defined compatibly, and simultaneously this proves that our scheme which configures the weight β according to the service type is true. In Fig. 6, the legend 'considered' notes the scheme considered service

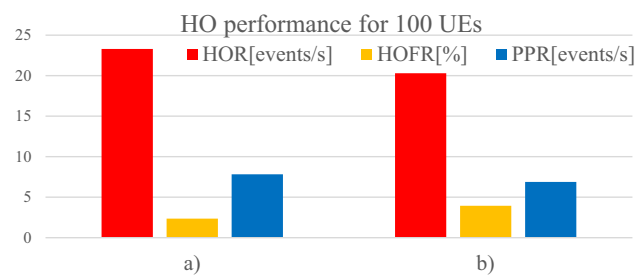

Fig. 5 The change in handover performance of network when the value of β is unvaried and varied according to the type of service for 100 UEs **a** HO performance of the network in the case of $\alpha = 0.8$, $\gamma = 0.1$, $\beta = 0.1$ for GBR and non-GBR service without considering service type of UE, **b** HO performance of the network in the case of $\alpha = 0.8$, $\gamma = 0.1$, $\beta = 0.1$ for GBR service and $\alpha = 0.8$, $\gamma = 0.1$, $\beta = 0.01$ for non-GBR service with considering service type of UE)

type, 'no-considered' notes not. Figure 6 shows that the enhancement of HO performance results in the energy saving.

Then, we estimated the optimum weights. In our work, we have defined $\beta = 0.1$ for the GBR service and $\beta = 0.01$ for non-GBR service as shown in Fig. 5, assuming that the RSRP and moving direction of UE does not impact on available bandwidth, thus even if α and γ are varied, the trend of β remains. This assumption has been proved through the simulation.

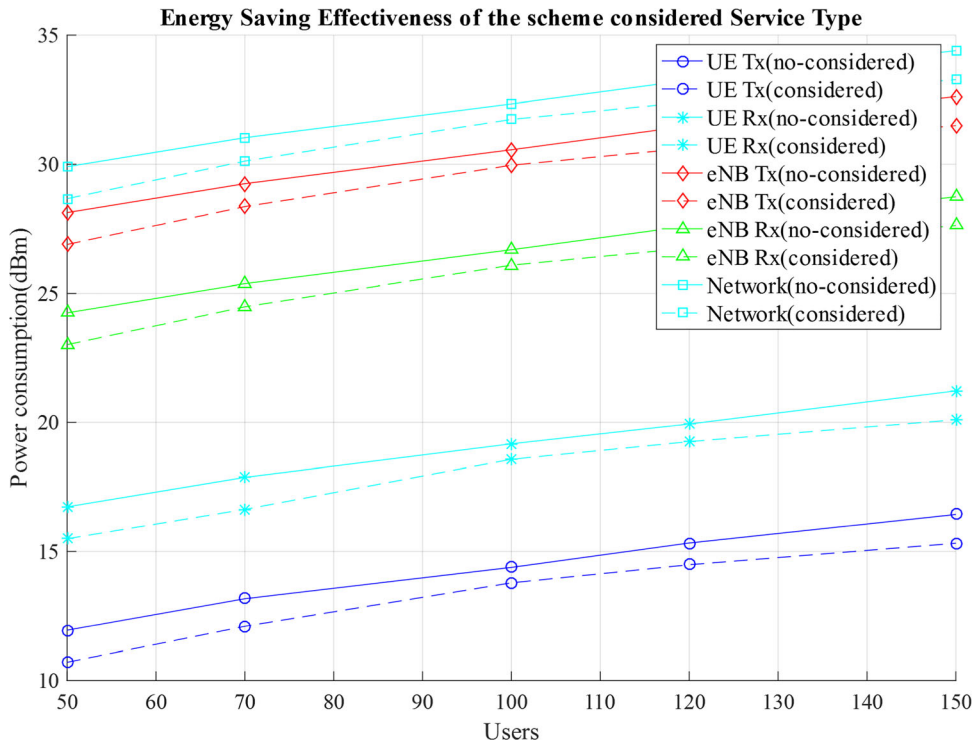

Fig. 6 The changes in power consumption of UE, eNodeB, and network with and without considering the service type. (Taking no account of service type, $\alpha = 0.8$, $\gamma = 0.1, \beta = 0.1$(GBR, non - GBR), taking account of service type, $\alpha = 0.8$, $\gamma = 0.1, \beta = 0.1$(GBR), 0.01(non - GBR))

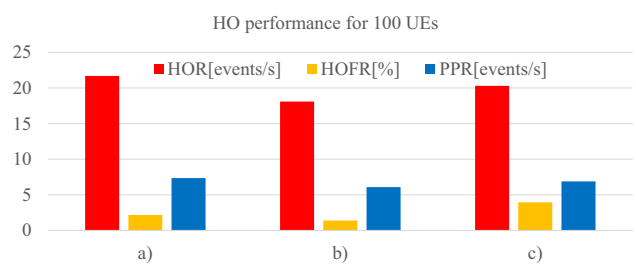

Fig. 7 HO performance comparison versus various α, γ and constant $\beta = 0.1$(in the case of GBR-service), 0.01(in the case of non-GBR service) **a** $\alpha = 0.6, \gamma = 0.3$, **b** $\alpha = 0.7, \gamma = 0.2$, **c** $\alpha = 0.8, \gamma = 0.1$)

Then, the performance where varying α and γ, satisfying the relationship $\alpha + \gamma = 1 - \beta$, is presented in Fig. 7. The power consumption comparisons with the different α and γ for both transmission and reception in terms of UE and BS side are given in Fig. 8, including power consumption comparison of network versus various parameters.

In the case of $\alpha = 0.7$, $\gamma = 0.2$, power consumption is the lowest due to low HOR and PPR as shown in Fig. 8(see legend 'optimized' in Fig. 8).

As a result of this, we have proposed that $\alpha = 0.7$, $\gamma = 0.2$, $\beta = \begin{cases} 0.1, & \textit{for GBR service} \\ 0.01, & \textit{for non-GBR service} \end{cases}$

5.3.2 Effectiveness evaluation of proposed scheme on the different indices

Based on defining the optimal weights, we have analyzed the effectiveness of the proposed scheme in the comparison of the previous schemes in terms of various characteristics including HO performance, power consumption, delay, PLR and throughput. The previous HO algorithms which are selected to compare are LABRBHA [27] based DL-HO, only uplink reference signal-based UL-HO, namely UL-RSRP schemes.

The comparison of the previous and proposed schemes is shown in Fig. 9. The proposed scheme shows the improvement of HOR, HOFR, and PPR due to the decrease of too early HOs, ping pong effect, frequent HOs because target cell is determined considering not only UL-RSRP but also available bandwidth of the BS and moving direction of UE.

The power consumption comparison (in detail) of the previous and proposed HO schemes is presented in Fig. 10. The signaling messages transmitted from the UE are MeasReport, RACH, HOconf and the message transmitted from the BS is HOcmd (In the figure, "Tx" means transmission and the "Rx" denotes reception). The signaling

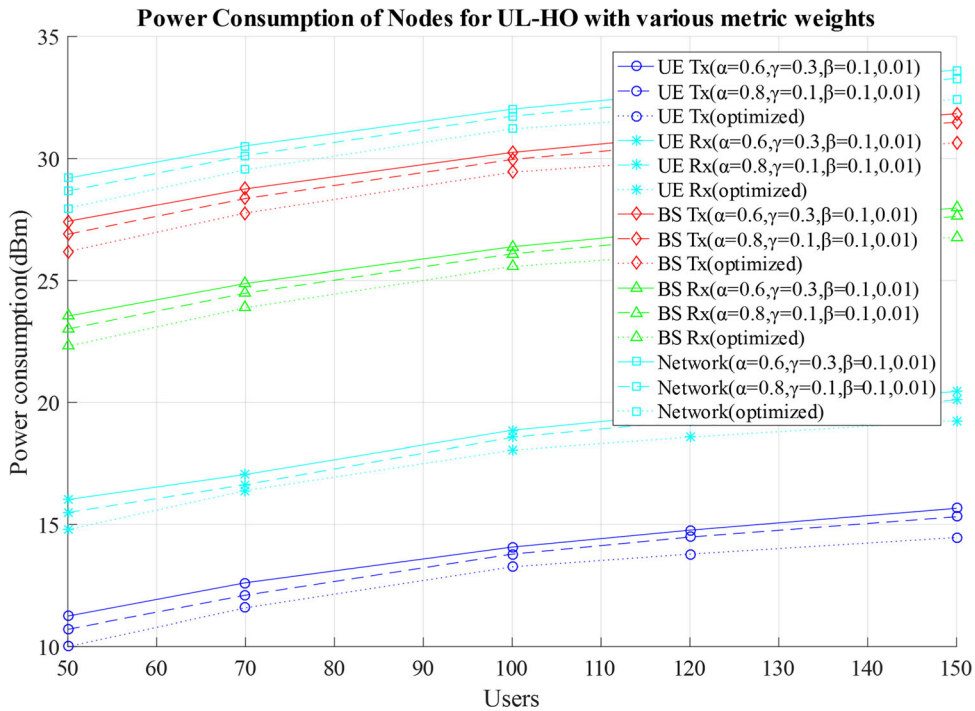

Fig. 8 Find optimal metric weights: Power consumption versus number of UEs for UE, eNodeB and Network with various α, γ (It is optimum when $\alpha = 0.7,\ \gamma = 0.2,\ \beta = 0.1(GBR),\ 0.01(non{-}GBR)$)

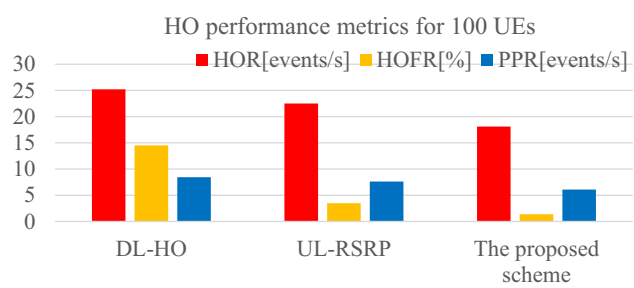

Fig. 9 HO performance comparison of the previous and proposed HO schemes

messages that are transmitted from the UE side are received in the BS and similarly, the messages transmitted from the BS side are received in the UE.

As shown in Fig. 10e, the proposed scheme saved energy of 40.5% than DL-HO, 20% than UL-RSRP.

Figure 11a presents the average delay of service per a UE under the different HO schemes. Here, the delay means not only the delay due to HO but also the total delay due to various reasons. However, all the methods except the HO scheme in all algorithms are the same, and this delay difference is mainly due to the difference of the HO scheme. Our algorithm has a gain of about 10 ms compared to UL-RSRP in terms of delay. Figure 11b shows that the proposed algorithm reduces PLR about 32% than UL-RSRP. As shown in Fig. 11c, the proposed algorithm increased the throughput about 2.6Mbps than UL-RSRP.

Finally, the simulation results shown in Fig. 9–11 are summarized in Table 4 for the numerical comparison between the proposed and the conventional schemes in terms of seven characteristics. The results show that the proposed scheme brings the high-performance improvement while reducing the power consumption than the previous schemes.

6 Conclusion

In this paper we have reconsidered the HO scheme which uses UL RS(SRS) during HO to improve the performance of HOs that occurs frequently in HetNet and proved that this scheme reduces the power consumption significantly. Based on this, we have introduced the concept of VPT, thus made the searching of the target cell more reasonable and faster in the network environment, HetNet where macro

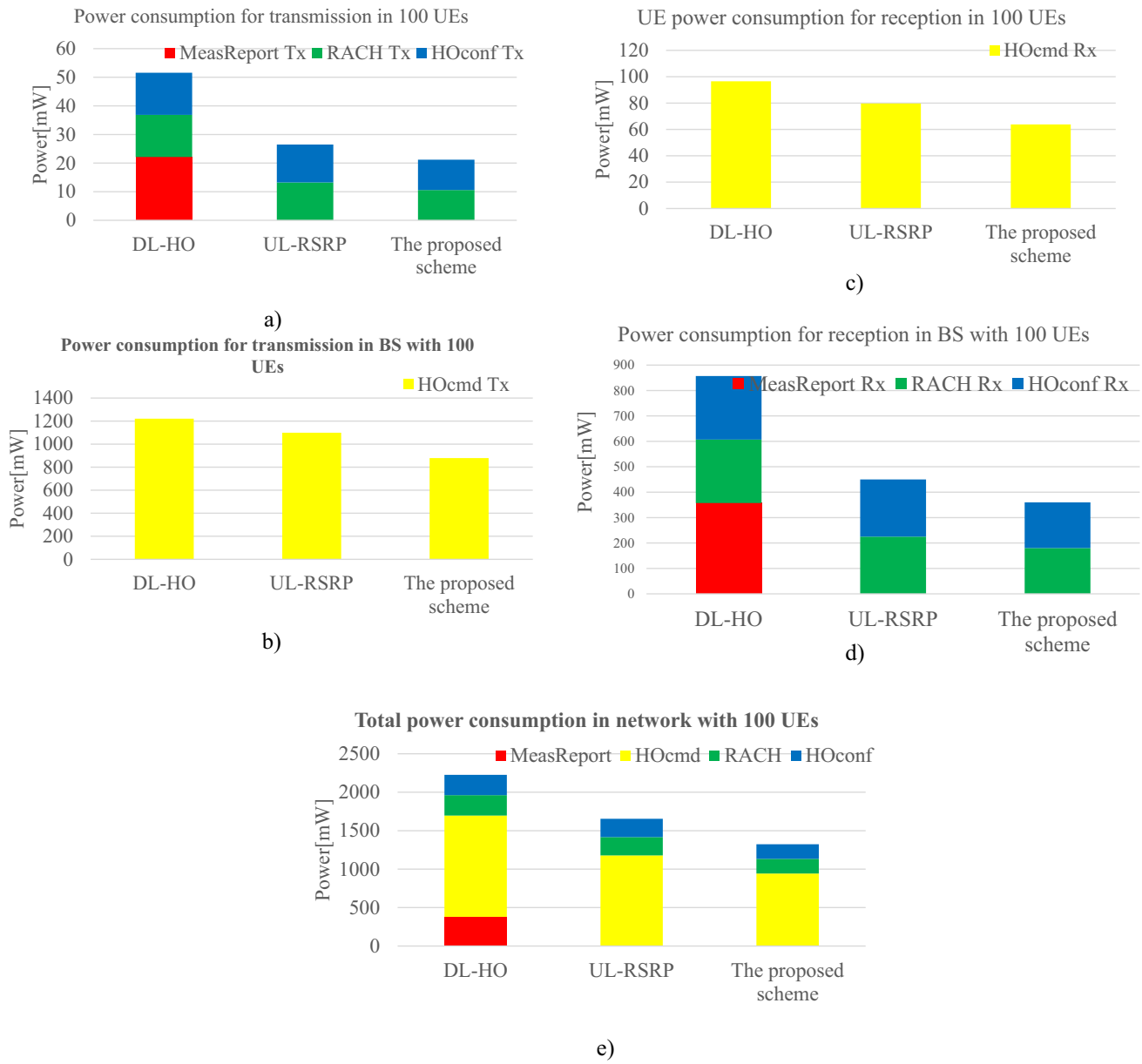

Fig. 10 Average power consumption comparison versus the various HO algorithms for transmission and reception in UEs, eNodeBs **a** Average transmission power consumption per a UE, **b** Average transmission power consumption per a BS with 100 UEs, **c** Average reception power consumption per a UE for HOcmd message from BS, **d** Average reception power consumption per a BS for various messages (including MeasReport, RACH and HOconf message) from UEs, **e** Total average power consumption in Network with 100 UEs)

cell coexists with small cells. In particular, we have proposed the new metric which considers UL RSRP, available bandwidth and moving direction of the UE. In this paper, we have also improved the HO performance by defining the weight of available bandwidth according to the service type. Besides, the weights are confirmed by simulation. The simulation results show that the proposed algorithm has the better performance than the traditional ones in terms of energy, HOR, HOFR, PPR, delay, PLR, throughput. The proposed algorithm is able to be used in Ultra-dense network (UDN) of LTE-A/5G NR. In the

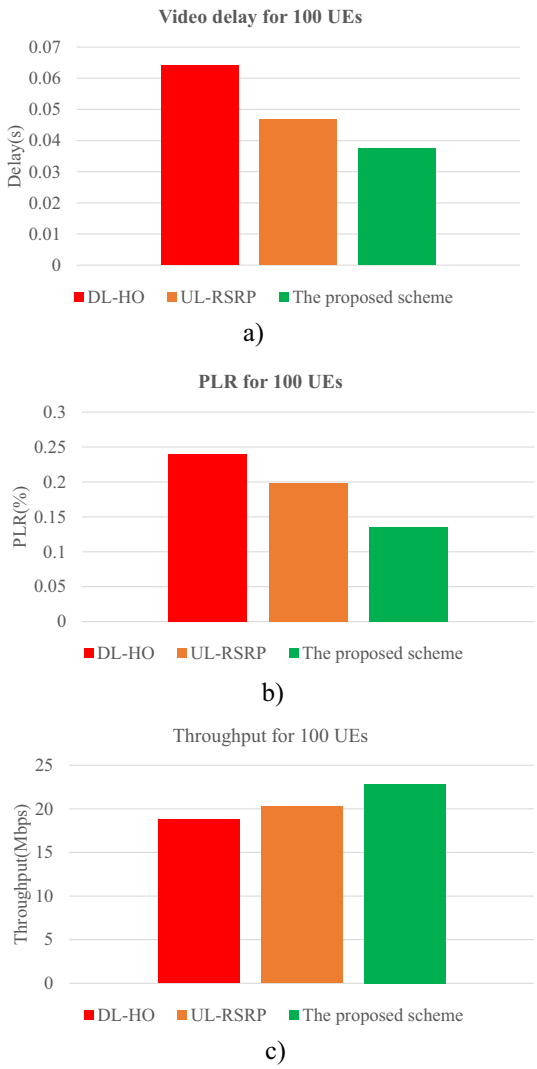

Fig. 11 QoS comparison of the various HO scheme in the network with 100 UEs for video streaming (**a** Average delay per a UE, **b** Average packet loss rate (PLR) per a UE, **c** Throughput)

Table 4 The performance parameters in various HO algorithms (for 100 UEs)

Performance parameters	DL-HO	UL-RSRP	The proposed algorithm
HOR (events/s)	25.2	22.5	18.1
HOFR (%)	14.51	3.48	1.39
PPR (events/s)	8.44	7.625	6.08
Total power consumption during HO in network (mW)	2226.2075	1655.0775	1324.062
Delay per a UE(s)	0.064	0.0467	0.0375
PLR per a UE (%)	0.24	0.1983	0.1356
Throughput (Mbps)	18.8	20.25	22.83

future works, we are going to study deeply on the procedures of UL-HO.

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Declarations

Conflict of interest The authors have no conflicts of interest to declare that are relevant to the content of this manuscript.

Availability of data and material Parameters in the experiments of this work are widely used in the previous research and practice, and comply with the standards of 3GPP and ITU-T. We assure that we will provide the experiment data as much as possible in case reviewers require the transparency of the experiment results of this work.

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